

Denis Crismariu

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Summary

Data Science and Artificial Intelligence master's student at TU/e with a background in Computer Science and Robotics, demonstrating strong proficiency in C++, C, Java, Python and Autodesk Fusion 360. Interested in AI/ML and low-level technologies, with a drive to explore innovative solutions and advance technical expertise.

Education

Eindhoven University of Technology (TU/e) **2026 - 2028**
Data Science and Artificial Intelligence Master

Alexandru Ioan Cuza University **2023 - 2026**
Computer Science Bachelor
Bachelor's thesis graded 10/10

Projects

Bachelor's Thesis: The Use of the NEAT Algorithm in Cooperative and Competitive Shooter Games
NEAT-based multi-agent training system in C++

- Built a complete **neuroevolution** system from scratch: a custom NEAT library, a 3v3 team-based shooter arena in raylib (walls, raycast sensors, bullet physics) and a **co-evolutionary** training framework in which agents learn to cooperate and compete from zero domain knowledge, without labeled data or a differentiable reward.
- Extended the library with defenses against topological collapse (output-node disconnection guard, asymmetric enable/disable mutation, output-bias clamping) and recurrent connections for per-agent memory; trained with role-separated populations, a dominance tournament with **binomial significance testing**, Hall-of-Fame sampling and curriculum phases.
- Main finding: reformulating the game's payoff structure (**King-of-the-Hill** with respawn) was a far greater lever for emergent behavior than refining the optimizer or the reward function.
- Documented a catalog of interventions that did not work (reward shaping, spawn curriculum, novelty search, memory scaling) as an attributable **negative result**.

NeuralNetworkCPP **July 2023 - November 2024**
Neural Network library in C++

Plug-and-play library exposing a small NeuralNetwork / DataSet / Gym API, with backpropagation, **SGD with momentum**, learning rate decay and **HogBatching**, a lock-free HOGWILD!-style **parallel mini-batch** update; raylib and stb are used for image handling and visualization. Ships with demos on MNIST subsets (image upscaling, digit-to-digit transition, a digit-compression network) plus adder and XOR examples.

Experience

ddroidd **September 2025 - August 2026**

Junior Software Engineer (January 2026 - August 2026) · AI Engineer Trainee (September 2025 - December 2025)

Bootstrap: project generator (Python, FastAPI, Jinja2, Docker)

- Built a CLI, interactive wizard and FastAPI REST endpoint that generate **ready-to-run, Docker Compose-based projects** for Python, Java (Spring Boot) and React (Next.js) stacks, with selectable database, message broker, cache, mailing and monitoring services, pinned container versions, and scaffolded tests and logging.
- Designed its **modular, catalog-driven architecture** around Jinja2 templates kept in a separate content repository, so any team can customize the templates or add a new language, service or container with minimal friction.

Ticketing middleware: issue forwarding to Desk365 (Node.js, Keycloak, Redis)

- Developed a Node.js (Fastify) service that forwards issues submitted in the company's core application to the Desk365 ticketing platform, automatically creating, assigning and **permission-scoping** tickets that were previously entered by hand.
- Secured it as a **multi-tenant** service with per-organization Keycloak realms, **JWT** verification and role-based Redis rate limiting, deployed with Docker Compose.

Skill Hub: self-hosted registry for AI agent skills (Python, FastAPI, Jinja2, Docker)

- Built an on-premise hub for hosting, discovering and sharing **AI agent skills** and curated skill collections (e.g. security packs), replacing ad hoc sharing over chat and repositories; skills are organized by tags with filtering, and a selection or collection can be bundled into a downloadable pack.
- Implemented versioning, comments and ratings, owner-only editing, and company-domain email verification for sign-in.

Skills

Languages: Python, C++, C, Java, JavaScript (Node.js)
ML & Data: PyTorch, TensorFlow, NumPy, pandas, scikit-learn, Matplotlib, Jupyter
Frameworks & Tools: FastAPI, React, React Native, Docker, Git, Linux, Fusion 360

Volunteering

CyLiis **September 2021 - Present**

Technical Team Leader / Mentor, *FIRST* Tech Challenge robotics team
cyliis

- Led the Programming and 3D Design departments, mentoring team members in coding and CAD tools.
- Organized and conducted training sessions, sharing knowledge in programming and 3D design.

Achievements include: Winner of the Maryland Tech Invitational 2024 and Semi-Finalist in 2023, 3rd Place at the Inspire Awards *FIRST* Tech Challenge Romanian Nationals, and Champion of the 1st and 2nd editions of the Bucharest Twin Cup.

Languages

Romanian (Native) **English (C1)**